MILITARY SPECIFICATION

SERVO, COLD GAS

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the Cold Gas Servo, critical item, referred to herein as the "servo."

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging and Packing Levels.

MILITARY

MIL-STD-109	Quality Assurance Terms and Definitions.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering Standardization Department (SESD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-810	Environmental Test Methods.
MIL-STD-1695	Environments, Working, Minimum Standards for.
MIL-STD-45662	Calibration Systems Requirements. ‘

2.1.2 Other Government documents, drawings, and publications. The following drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

639AS3485	Housing, Guidance Control Section.
639AS3957	Base, Tank.
639AS6360	Servo Assembly; Cold Gas.
639AS6460	Pressure Vessel Assembly.
639AS6491	Shipping Container, Servo Assembly.
SA 492551	SW IC Vibration Test Fixture Base Hole Pattern.

PUBLICATIONS

Naval Air Systems Command
(Code Ident 30003)

AD-1322	Reliability Program Provisions for Production of Sidewinder AIM-9.
WS-6536	Procedures and Requirements for Preparation and Soldering of Electrical Connections.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS.

3.1 Item description. The servo is an electropneumatic mechanical device that has self-contained energy in the form of compressed gas and a thermal battery that will, on command, provide missile electrical power and torque to two axes of missile-control surfaces proportional to the electronic control commands.

3.2 Characteristics.

3.2.1 Performance. The servo shall meet the following performance requirements.

3.2.1.1 Stall torque. The servo stall torque shall be greater than 1,520 inch-pounds (in-lb) in both directions for each axis while restrained at 84 ±2 in-lb per-degree coaxial load (see Figure 1). The stall torque in both directions for a given axis shall be within 100 in-lb.

3.2.1.2 Torque hysteresis. The torque hysteresis shall be within the boundaries of Figure 1 while following a triangular electronic control signal. Servo hysteresis shall be less than 200 in-lb between ±13.0 V and less than 240 in-lb at all other control input voltages. The torque hysteresis shall be greater than 50 in-lb between ±14.5 volts (see Figure 1).

3.2.1.3 Torque symmetry. The symmetry of the torque hysteresis trace shall be within 75 in-lb where the trace crosses ±500 in-lb and within 100 in-lb where the trace crosses 1,050 in-lb.

3.2.1.4 Torque unbalance. With no electronic control input, the servo shall have less than 75 in-lb torque per axis between 0.220 and 0.720 seconds following application of 1,105 ±15 pounds per square inch gage (psig) pressure to the ground test port.

3.2.1.5 Step response. The servo torque output shall be 63 percent of a ±560 ±40 in-lb steady state output within 120 to 180 milliseconds (ins) from the input command signal. At low temperature (operating), the time shall be within 120 to 210 ms.

3.2.1.6 Crosstalk. The servo crosstalk (cross axes torque drop) shall be less than 40 in-lb when a full torque (not less than 1,520 in-lb) step input is achieved at both axes with one axis lagging or leading by 0.5 ±0.1 second (see Figure 2).

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3.2.1.7 Acceleration sensitivity. With no electronic command input and 1,105 ±15 psig gas pressure applied to the ground test port, the servo shall produce torque outputs less than the following values at the stated condition (see Figure 3).

a.	Forward acceleration:	25 ±1 gravity units (g)	25 (in-lb)
b.	Aft acceleration:	10 ±1 g	100 (in-lb)
c.	Combined axis acceleration:	40 ±1 g	100 (in-lb)

3.2.2 Physical characteristics. Physical characteristics shall be in accordance with Drawing 639AS6360.

3.2.3 Reliability program. When specified in the contract or purchase order, a reliability program shall be planned, implemented, conducted, and reported in accordance with AD-1322 (see 6.2.2).

3.2.3.1 Reliability demonstration. When a first article sample is specified in the contract or purchase order (see 6.2.3), reliability of 0.91 at the lower confidence limit of 80 percent will be demonstrated by successfully completing the first article inspection specified herein with no failures. Subsequent to first article inspection and acceptance, product quality shall be maintained and verified commensurate with the quality conformance inspection of 4.2. The inherent design reliability shall be maintained throughout the product manufacturing sequence. A periodic assessment of cumulative quality conformance inspection data, shall be performed to verify that product quality and reliability has not been degraded.

3.2.4 Environmental conditions. The servo shall meet the requirements of 3.2.1 when subjected to temperatures ranging from +74 ±3 degrees Celsius (°C) to -48 ±3°C, and after being subjected to the environments specified herein.

3.2.4.1 Environmental (nondestructive). The servo shall meet the requirements of 3.2.1 either during or subsequent to the following conditions:

a. Shock, Figure 4 (see 4.8.1).

b. Vibration (random), Figure 5 (see 4.8.2).

c. Temperature shock (see 4.8.3).

d. High temperature (operating) (see 4.8.4).

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e. Low temperature (operating) (see 4.8.5).

f. Humidity (see 4.8.7).

3.2.4.2 Duty cycle and readiness time. The servo shall contain sufficient energy to perform the flight profile shown in Table I with both axes operating while restrained with 84 ±2 inch-pound per degree (in-lb/deg) coaxial loading. One axis shall lag the run time shown in Table I by 0.5 ±1 second. The-servo shall accept an external electrical signal of 2.8 amps to the electro-initiator and provide a torque output of not less than 900 in-lb within 0.8 second. The electro-initiator shall function within 50 ms.

TABLE I. Duty cycle (leading axis).

Run time 1/ (±0.1 sec)	Signal volts peak (±0.1 volts)	Torque 2/ (in-lbs)	Frequency (Hz)	Wave shape
0.4	+21.6	1,520 min	---	Step
1.9	-21.6	1,520 min	---	Step
3.4	±1.2	---	0.5	Sine
25.4	±3.1	---	0.5	Sine
27.4	±7.4	---	0.5	Sine
29.4	-14.2	1,000 min	---	Step
30.9	+14.2	1,000 min	---	Step
32.4	±1.2	---	0.5	Sine
38.4	+21.6	1,520 min	---	Step
39.5	+21.6	1,520 min	---	---

1/ Actuation at 0.0 second.

2/ The torque specified shall be met during any portion of the step input.

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3.3 Design and construction. The servo shall be manufactured in accordance with Drawing 639AS6360.

3.3.1 Certification. When specified in the contract or purchase order (see 6.2.2), the contractor shall prepare written certification accompanied by objective quality evidence, in accordance with MIL-STD-109, that the materials, processes, and parts used in the servo meet the requirements of this specification.

3.4 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspections (see 4.1.2 and 6.2.3).

3.5 Workmanship. The servo shall be constructed and finished in a manner to ensure compliance with the requirements of this specification and MIL-STD-454, Requirement 9. Particular attention shall be paid to marking of parts and assemblies, wiring, plating, machine screw assemblies, and freedom of parts from burrs, sharp edges, or any other damage or defect that could make the part unsatisfactory for the purpose intended.

3.5.1 Special working conditions. The contractor shall provide adequate facilities for the fabrication, assembly, and testing of servos to be delivered in accordance with this specification. Unless otherwise specified in the contract or purchase order (see 6.2.1), the minimum standards for working environments of MIL-STD-1695 shall apply.

3.5.2 Soldering. Procedures and requirements for preparation and soldering of electrical connections shall be in accordance with WS-6536.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order (see 6.2.1), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of inspections.

a. First article inspections (see 4.1.2).

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b. Quality conformance inspections (see 4.2).

c. Reliability inspections (see 4.7).

4.1.2 First article inspections. When specified in the contract or purchase order (see 6.2.3), a first article sample of 18 servos shall be delivered to the testing activity designated in the contract or purchase order. The first article sample shall be manufactured using the same methods, materials, processes, and procedures proposed for production. Any production prior to acceptance of the first article sample is at the risk of the contractor.

4.1.2.1 First article inspection sequence. The first article sample shall be subjected to the tests and inspections of Table II. Prior to presentation of the first article sample for acceptance by the cognizant Government representative, the contractor shall maintain objective quality evidence, in accordance with MIL-STD-109, that the drawing, product, and performance requirements specified herein have been met for each servo.

4.1.2.2 Captive carry environment inspection. A captive carry environment inspection shall be conducted in accordance with the following:

a. A test sample of six servos shall be selected from those servos in the first article sample that have passed the inspections of Table II, Group A.

b. Prior to the captive carry environment inspection, the servo shall be in accordance with Drawing 639AS6360. The pressure vessel weight shall be verified and recorded for each pressure vessel.

c. Environmental exposure, test profile, and test duration shall be in accordance with the 8-hour test cycle shown on Figure 6 while mounted into a test fixture which simulates a guidance control section.

d. At the conclusion of every 6 ±3 test cycles, as shown on Figure 6, the servo shall be tested in accordance with the inspections of Table II, Group A.

e. At the conclusion of testing, the pressure vessel shall be weighed and the pretest and post-test weights compared. A variation of 0.1 grams or greater (post-test weight less pretest weight) shall be considered a failure of the first article sample. Accuracy of the measurement shall be within 0.01 gram.

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f. When specified in the contract or purchase order (see 6.2.2), a failure summary and analysis report shall be prepared for all failures.

4.1.2.3 First article acceptance criteria. Failure of any servo to meet the requirements specified herein shall cause rejection of the first article sample.

4.2 Quality conformance inspections. Each servo produced shall be subjected to the inspections of Table II, Group A. Failure of any servo to meet any of the requirements specified herein shall cause rejection of the servo.

TABLE II. First article and quality conformance inpections.

Inspections	Requirement paragraph	Method paragraph
Group A - 100 percent (performance)		
Visual and mechanical	3.2.2., 3.3, and 3.5	4.6.1
Torque characteristics		
Stall torque	3.2.1.1	4.6.2.1
Torque hysteresis	3.2.1.2	4.6.2.1
Torque symmetry	3.2.1.3	4.6.2.1.1
Torque unbalance	3.2.1.4	4.6.2.2
Step response	3.2.1.5	4.6.2.3
Group B - Four servos (environmental nondestructive)		
Shock	3.2.4.1a	4.8.1
Vibration (random)	3.2.4.1b	4.8.2
Temperature shock	3.2.4.1c	4.8.3
High temperature (operating)	3.2.4.1d	4.8.4
Low temperature (operating)	3.2.4.1e	4.8.5
Crosstalk	3.2.1.6	4.6.2.4
Acceleration sensitivity	3.2.1.7	4.6.2.5
Duty cycle and readiness time [1/]	3.2.4.2	4.8.6
Humidity [1/]	3.2.4.1f	4.8.7
Group C - Six servos		
Captive carry environmental	3.4	4.1.2.2

1/ Tests shall be performed in the sequence shown and after all preceding tests.

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4.2.1 Lot acceptance tests. A random sample consisting of five percent of the lot but not less than two servos (see 6.3.3) that have been subjected to and passed the quality conformance inspections of 4.2 shall be subjected to and pass the following environmental inspections of Table II Group B:

a. Shock .

b. Vibration (random).

c. Temperature shock.

d. High temperature (operating).

e. Low temperature (operating).

The servos used for lot acceptance testing are to be refurbished and delivered after completion of the above tests and quality conformance testing.

4.2.1.1 Lot acceptance criteria. Failure of any servo to meet the requirements of 4.2.1 shall cause rejection of the lot (see 6.3.3).

4.3 Test equipment and conditions. Unless otherwise specified in the contract or purchase order (see 6.2.1), test equipment, facilities, calibration procedures, and test procedures shall be subject to approval by the procuring activity prior to Government acceptance of produced items. The contractor shall furnish and maintain all necessary test equipment, facilities, and personnel for the performance of all contractor-performed tests. The contractor shall provide a minimum of 1 week advance notice of his readiness for inspection of test equipment, test procedures, and facilities. The contractor shall have a report available for the use of inspecting personnel, which analyzes input and output functional measurement accuracy attainable at each test position and compares this accuracy with the accuracy required by the documents and drawings.

4.3.1 Test equipment accuracy. Unless otherwise specified in the contract or purchase order (see 6.2.1), all test equipment shall have an accuracy of at least 15 percent of the maximum allowable tolerance range when both upper and lower limits are given, or 1 percent of the value being measured when only one limit (either upper or lower) is given. All electrical and electronic measuring equipment used in acceptance testing and mechanical assembly shall be calibrated in accordance with a system meeting the requirements of MIL-STD-45662. The contractor shall certify that all test equipment has been so calibrated before acceptance testing begins. Recalibration shall be made at necessary intervals or as required by the Government representative.

4.3.2 Servo torque measuring equipment. The servo torque measuring equipment shall be of such a nature and design that the accuracy of the reading is within ±1 percent or 2 in-lb, whichever is greater.

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4.4 Failure reporting, analysis, and corrective action. The contractor shall perform failure reporting, failure analysis, and corrective action in accordance with AD-1322 and as specified in the contract or purchase order (see 6.2.1). Corrective action for a design, part, or material related failure s-hall require approval by the procuring activity.

4.4.1 Relevant failures. A relevant failure is any failure except those classified and listed herein as a nonrelevant failure. Concurrence by the procuring activity shall be required for classification of all failures experienced during testing.

4.4.2 Nonrelevant failure.

a. Any failure caused by test equipment.

b. Any failure that results from operational or procedural error in testing.

c. Secondary failures when they are confirmed to be induced by failure of another part. At least one primary (relevant) failure shall be charged when a secondary failure is claimed.

d. When a failure results from a design incompatibility which has been determined to exist and concurred with in writing by the procuring activity, provided that the required corrective action for the design incompatibility does not apply to the remainder of the servos.

'e. Repeated failures for which corrective action has been documented but, by written direction of the procuring activity, the corrective action is not incorporated into the remainder of the servos.

4.5 Test conditions and provisions.

4.5.1 Test sequence. Unless otherwise specified herein, the contractor shall perform testing in any sequence that will best utilize facilities and meet schedules while meeting all 'requirements specified herein.

4.5.2 Test plans. When specified in the contract or purchase order (see 6.2.2), detailed test plans shall be prepared by the contractor and shall be subject to approval by the procuring activity prior to use. Actual performance values measured shall be recorded.

4.5.3 Test reports. When specified in the contract or purchase order (see 6.2.2), test reports shall be prepared by the contractor. Distribution of these test reports shall be as specified in the contract or purchase order (see 6.2.1).

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4.5.4 Test conditions. Unless otherwise specified herein or in the applicable drawings, all measurements shall be made under existing ambient environmental conditions at the test facility.

4.5.5 Test station certification. Unless otherwise specified in the contract or purchase order (see 6.2.1), all test stations used for measuring requirements of the servo shall require certification by the procuring activity.

4.5.6 Warm-up time for electronic recording devices. All electronic recording devices used during performance of the electrical tests shall be allowed to warm-up and stabilize prior to performing these tests.

4.6 Test methods.

4.6.1 Visual and mechanical inspection. The servo shall be visually examined to verify conformance with 3.2.2, 3.3 and 3.5. The servo shall conform to the requirements of Drawing 639AS6360.

4.6.2 Performance test. The servo shall be mounted in a test fixture which restrains each axis with a coaxial load of 84 ±2 in-lb/deg and shall be tested as follows to verify the performance requirements of Table II, Group A. All tests except step response may be performed using air or nitrogen. The step response test shall be performed using a mixture by volume of 10 ±l percent helium and 90 ±1 percent air or nitrogen. Gas used for all testing shall be supplied to the ground test port at 1,105 ±15 psig with a dew point not less than -60°C and shall be filtered to 20 microns.

4.6.2.1 Torque characteristics. The electronic control input shall be a triangular wave signal with a peak amplitude of ±25 ±0.1 Vdc and a sweep rate of 1.11 ±0.01 volts per second. When tested under these conditions, the servo shall meet the requirements of 3.2.1.1, 3.2.1.2, and 3.2.1.3.

4.6.2.1.1 Torque symmetry. Torque symmetry shall be determined as follows (see Figure 7):

a. The voltage at which the center of each hysteresis curve measured vertically crosses the plus 500 in-lb line shall be determined. This voltage shall be recorded as voltage V1.

b. The voltage at which the center of each hysteresis curve crosses the zero in-lb line shall be determined. This voltage shall be recorded as voltage V0.

c. The parameter V2 shall be calculated in accordance with the following formula:

$$V2 = 2V0 - V1$$

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d. The torque value corresponding to the center of the hysteresis curve and V2 shall be determined. The torque at voltage V2 shall be within 75 in-lb of the torque at voltage V1.

e. The voltage at which the center of each hysteresis curve measured vertically crosses the plus 1,050 in-lb line shall be determined. This voltage shall be recorded as voltage V3.

f. The parameter V4 shall be calculated in accordance with the following formula:

$$V4 = 2V0 - V3$$

The torque value corresponding to the center of the hysteresis curve and V4 shall be determined. The torque at voltage V4 shall be within 100 in-lb of the torque at voltage V3.

4.6.2.2 Torque unbalance. Air or nitrogen shall be applied to the servo in less than 0.05 second and the torque measured for each of five applications. The servo shall meet the requirements of 3.2.1.4.

4.6.2.3 Step response. The electronic control input to each axis simultaneously shall be a square wave signal adjusted to produce a servo steady state output of ±560 ±40 in-lb. The step response shall be calculated for not less than five complete cycles and shall meet the requirement of 3.2.1.5 for each cycle.

4.6.2.4 Crosstalk. The electronic control input to each axis independently shall be a square wave adjusted to produce a servo output of not less than ±1,520 in-lb torque in each axis. The first input signal shall lag the second input signal by 0.5 ±0.1 second for five cycles then lead the second input signal by 0.5 ±0.1 second for five cycles. The servo shall meet the requirement of 3.2.1.6.

4.6.2.5 Acceleration sensitivity. The servo shall be mounted in a fixture capable of applying accelerations individually as shown in Figure 3. With zero volts command input, the torque output of each axis of the servo shall be recorded. The torque output shall be as specified in 3.2.1.7.

4.7 Reliability. When specified in the contract or purchase order (see 6.2.1), reliability demonstration tests shall be performed in accordance with a test plan prepared by the contractor and approved by the procuring activity.

4.8 Environmental (nondestructive) tests. Unless otherwise specified herein, when tests are conducted at high and low temperatures, the servo shall be stabilized (see 6.3.2) at the specified temperature. If any servo is removed from the temperature chamber during any portion of the following

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tests, that test shall be completed within 15 minutes of removal of each servo from the chamber. If any test is not completed within the 15-minute period, the servo shall be restabilized before continuing the tests. Unless otherwise specified herein, the servo shall be allowed to stabilize at room ambient conditions prior to performance of post tests and measurements. Upon completion of each of the environmental tests specified herein, the servo shall be inspected for evidence of damage and deterioration due to exposure to the environments specified.

4.8.1 Shock. The shock test shall be performed as follows:

a. The servo shall be mounted in a guidance control section housing (see Drawing 639AS3485) containing a tank base (see Drawing 639AS3957 or equal) at the servo and housing interface. The servo and tank base shall be mounted to the housing using screws NAS 1189E3P7H torqued to 25 to 30 inch pounds. The assembled unit shall be mounted in the test fixture (see Drawing SA 492551). Shock pulses shall be directed along each of the three mutually perpendicular axes of the test unit. Al 1 transverse shock tests shall have the acceleration transducer mounted on the fixture within 1 inch of the combined actuator and fixture center of gravity. When the servo is subjected to the shock test along its longitudinal axis, the acceleration transducer shall be mounted in the base of the shock table. The servo shall be nonoperating during the shock test.

b. The servo shall be subjected to two shock pulses, 47.5 ±2.5 g peak amplitude, 21 ±3 ms duration (see Figure 4) half-sine pulse in each of the six directions along the three mutually perpendicular axes for a total of 12 shock pulses. At the conclusion of shock tests, the servo shall be tested for and shall meet the requirements of the torque characteristics, torque unbalance, and step response tests of Table II.

4.8.2 Vibration (random). The servo shall be mounted as in 4.8.1.a. The vibration tests shall consist of ambient temperature vibration, low temperature vibration (-48 ±3°C) and high temperature vibration (+74 ±3°C). Provisions for conducting these tests shall include instrumentation that is calibrated at the temperature extremes required for the test. The tests shall be performed as follows:

a. The servo shall be attached to a vibration table for vibration in each of the three mutually perpendicular axes (see Figure 8). Vibration excitation along each axis shall be applied to the table with the geometric center of the fixture parallel with the vibration exciter armature centerline.

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b. The servo shall be subjected to random vibration as specified in Figure 5 for 30 minutes on each of the three axes at each temperature. During the ambient temperature vibration (Figure 5(c)), the servo shall be operated for sufficient time to verify the hysteresis, torque symmetry, and torque unbalance requirements of 3.2.1. Testing shall conform to the requirements of 4.8 except that the control input voltage shall be limited to 10 Vdc. At the conclusion of the vibration tests, the servo shall be tested for and shall meet the requirements of the torque characteristics, torque unbalance, and step response tests of Table II.

4.8.3 Temperature shock. The servo shall be subjected to the temperature shock tests conducted in accordance with the following modification of MIL-STD-810, Method 503.1 (see 6.4). The servo shall be nonoperating during the test. The test shall be conducted as follows:

a. Step 1. The servo shall be placed in a suitable high temperature chamber with an internal temperature of +74 ±3°C and maintained at this temperature for not less than 2 hours.

b. Step 2. The servo shall be transferred within 5 minutes to a cold chamber with an internal temperature of -54 ±3°C and maintained at this temperature for not less than 2 hours.

c. Step 3. The servo shall be returned to the high temperature chamber within 5 minutes after removal from the low temperature chamber.

Steps 1 through 3 shall constitute one cycle of temperature shock testing. The servo shall be subjected to five cycles of testing and, at the conclusion of the fifth 2-hour cooling period of Step 2, the servo shall be removed from the test chamber and allowed to return to room ambient temperature. The servo shall be tested for and shall meet the requirements of the torque characteristics, torque unbalance, and step response tests of Table II.

4.8.4 High temperature (operating). The servo shall be tested to the torque characteristics, torque unbalance, and step response tests of Table II at a temperature of 74 ±3°C. Gas at a temperature of 74 ±3°C shall be supplied to the servo for these tests. Gas temperature instrumentation shall not be required if the gas supply is stablized in the same environmental chamber with the servo.

4.8.5 Low temperature (operating). The servo shall be tested to the torque characteristics, torque unbalance, and step response tests of Table 11 at a temperature of -48 ±3°C. Gas at a temperature of -48 ±3°C

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and a dew point less than -60°C shall be supplied to the servo for these tests. Gas temperature instrumentation shall not be required if the gas supply is stabilized in the same environmental chamber with the servo.

4.8.6 Duty cycle and readiness time. To preclude activating the thermal battery, disconnect wire No. 2 from filter FL 1101 (see Drawing 639AS6360). Use wire No. 2 and servo ground to fire the electro-initiator. The servo shall be stabilized to a temperature of -48 ±3°C. The servo shall meet the requirements of 3.2.4.2. The test shall be initiated by supplying a firing current of 2.8 ±0.1 amps to release the gas stored in the servo pressure vessel .

4.8.7 Humidity. The servo, mounted in a test shell with simulated mating interfaces in place, shall be exposed to humidity in accordance with MIL-STD-810, Method 507.1, Procedure I (see 6.4). At the conclusion of the humidity test, the servo shall be tested for and shall meet the requirements of the torque characteristics, torque unbalance and step response tests of Table II. The pressure vessel is not charged during this test.

4.9 Preparation for shipment. Each servo, packaged and packed for shipment or storage, shall be visually inspected to verify conformance with the requirements of Section 5.

5. PACKAGING.

5.1 Preservation, packaging. Preservation and packaging shall be in accordance with FED-STD-102, Level A.

5,1.1 Level A. Servos shall be packaged in accordance with Drawing 639AS6491.

5.2 Packing. Packing shall be in accordance with FED-STD-102, Level A.

5.2.1 Level A. Servos shall be packed in accordance with Drawing 639AS6491.

5.3 Marking. Containers shall be marked in accordance with Drawing 639AS6491.

6. NOTES.

6.1 Intended use. The servo is intended for use as a fin actuation system in the Sidewinders AIM-9M Guided Missile.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Special working conditions (see 3.5.1).

c. Responsibility for inspection and inspection facilities (see 4.1).

d. Test equipment and conditions (see 4.3).

e. Test equipment accuracies (see 4.3.1).

f. Special test equipment (see 4.3.2).

g. Failure reporting, analysis, and corrective action (see 4.4).

h. Test report distribution (see 4.5.3).

i. Test station certification (see 4.5.5).

j. Reliability demonstration testing, if required (see 4.7).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data" requirements identified below should be developed as specified by an approved Data Item Description (DD From 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 7-104.9(n) (2) are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph	Data requirements	Applicable DID	Options
3.2.3	Reliability program plan	DI-R-7079	---
3.3.1	Certification	UDI-A-23264B	---
4.1.2.2f.	Failure Summary and Analysis Report	DI-R-7041	---
4.5.2	Test plans	DI-T-5204	---
4.5.3	Test reports	DI-T-2072	---

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(Data item descriptions relative to this specification, and identified in Section 6, will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.3 First article. When a first article inspection is required, the servo will be tested and should be a first article sample. The first article should consist of 18 servos as specified in 4.1.2. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, approval, and disposition of the first article.

6.3 Definitions.

6.3.1 Operating condition. Operating condition is defined as conformance of the servo output performance to the requirements specified in 3.2.1.

6.3.2 Stabilization (temperature). Stabilized temperature is defined as a condition where the surface temperature of the item (measured on the major mass of the system at not less than two places) does not vary more than 3°C from the regulated environmental temperature over a period of not less than 15 minutes. Once stabilization times have been established for the test conditions, those times may be used with written approval of the procuring activity.

6.3.3 Lot definition. For purposes of lot acceptance a lot shall be defined as one month's uninterrupted production of servos.

6.4 Environmental testing. MIL-STD-810C should be used when performing the environmental tests of 4.8.3 and 4.8.7.

6.5 Safety precautions. The loading, assembly, and handling of the explosives, subassemblies, and the finished items covered by this specification, involve hazardous operations and therefore require explosives safety precautions. Standard safety precautions for explosive-loaded items are contained in DoD 4145.26M.

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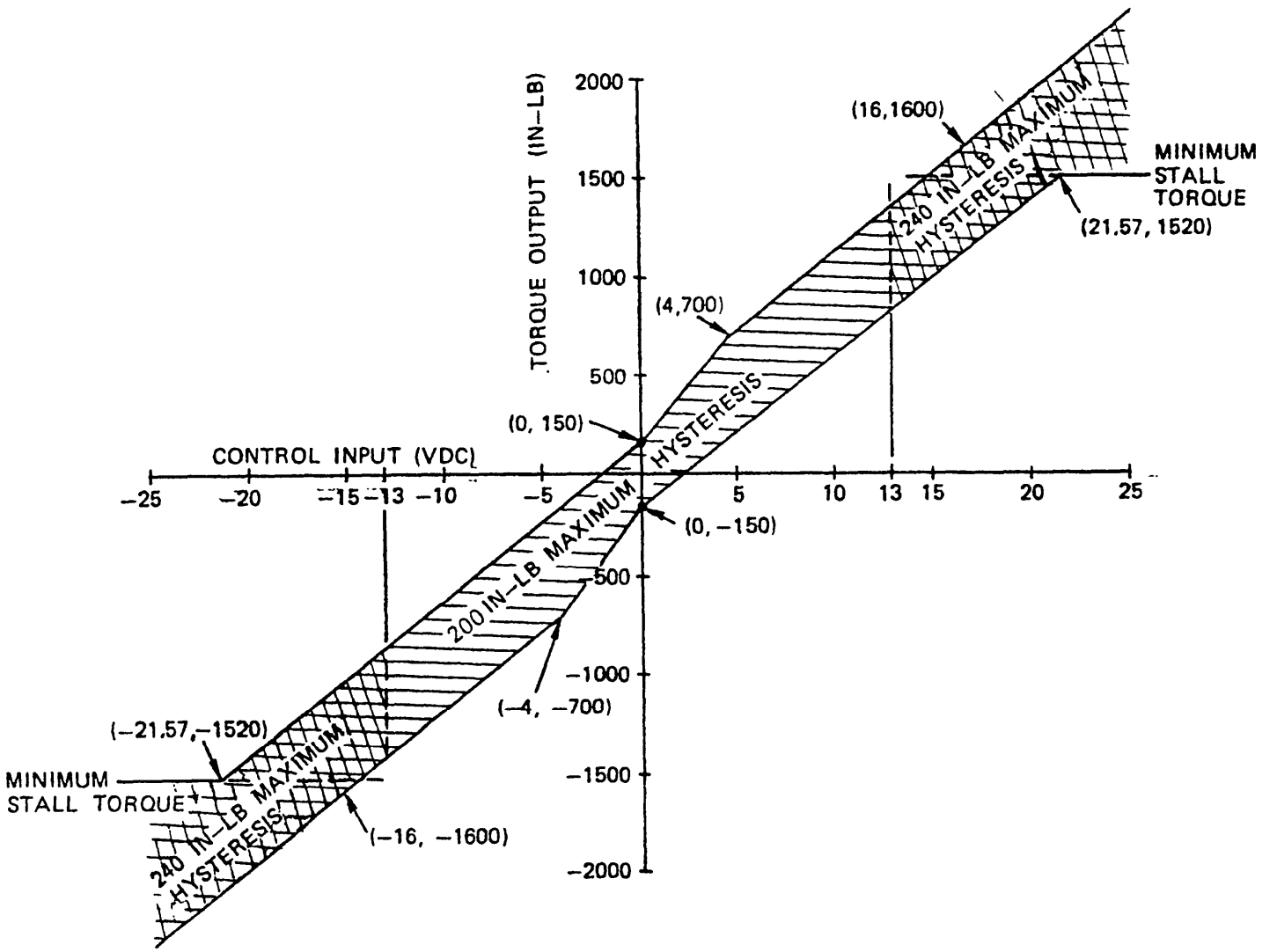


FIGURE 1. Torque/voltage performance envelope.

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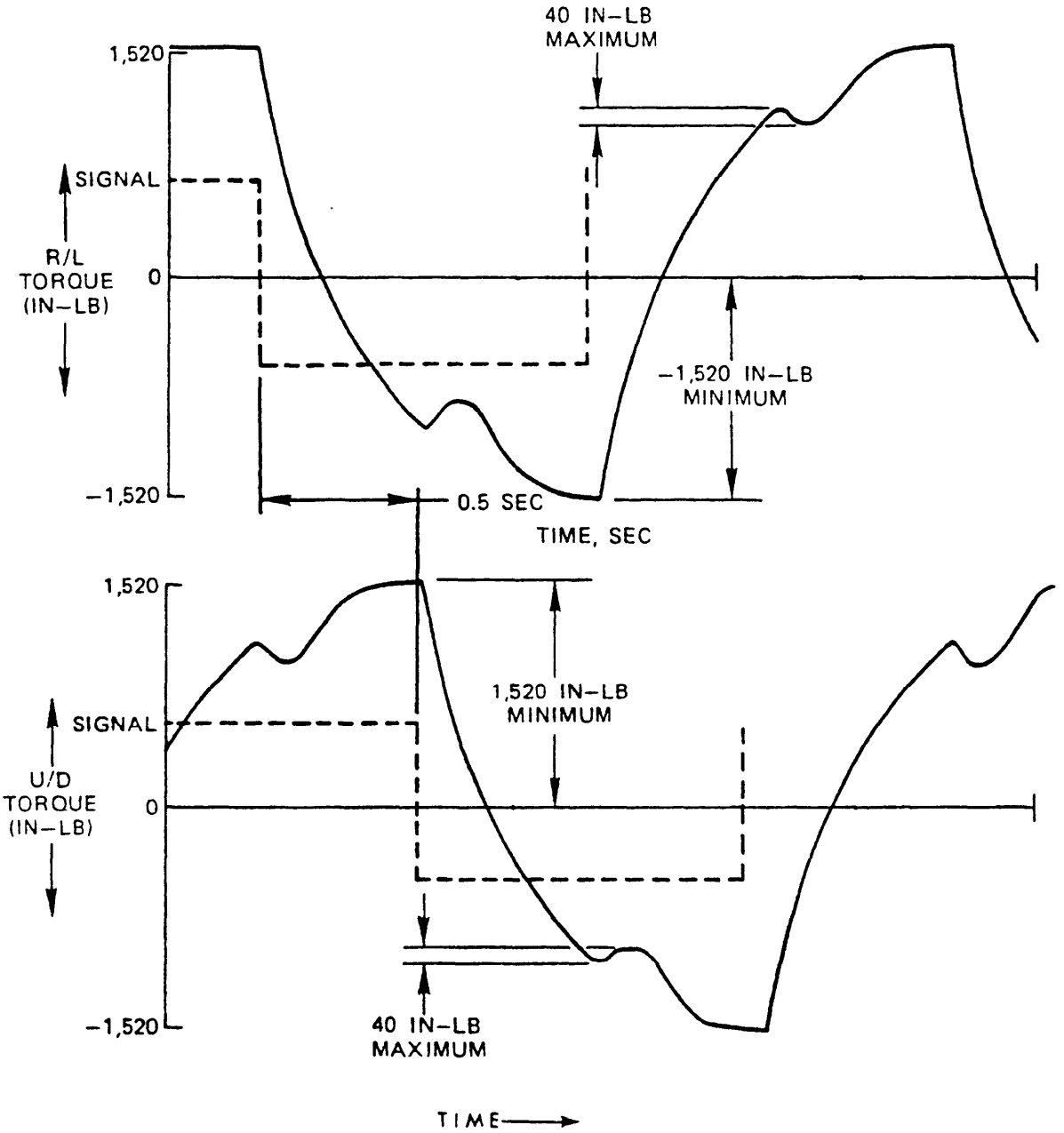


FIGURE 2. Crosstalk.

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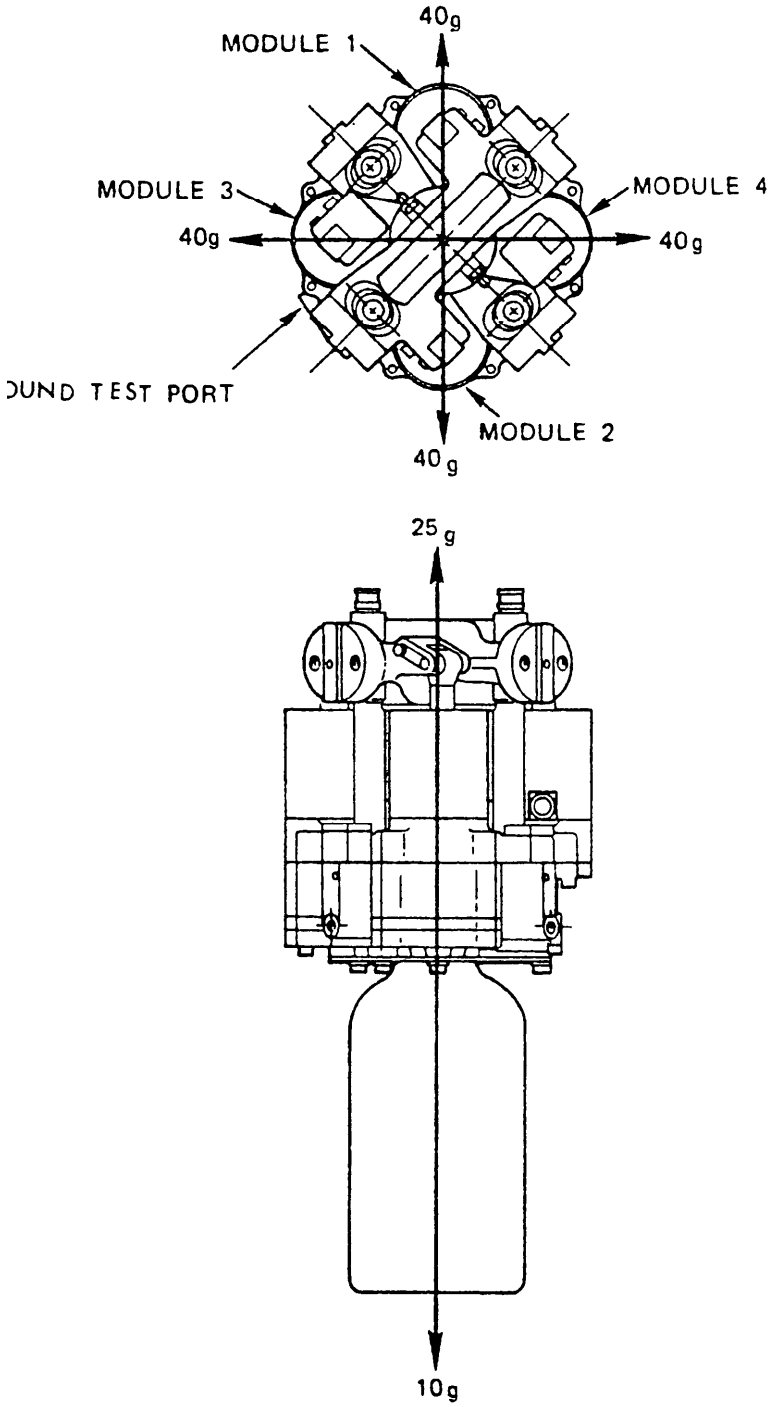
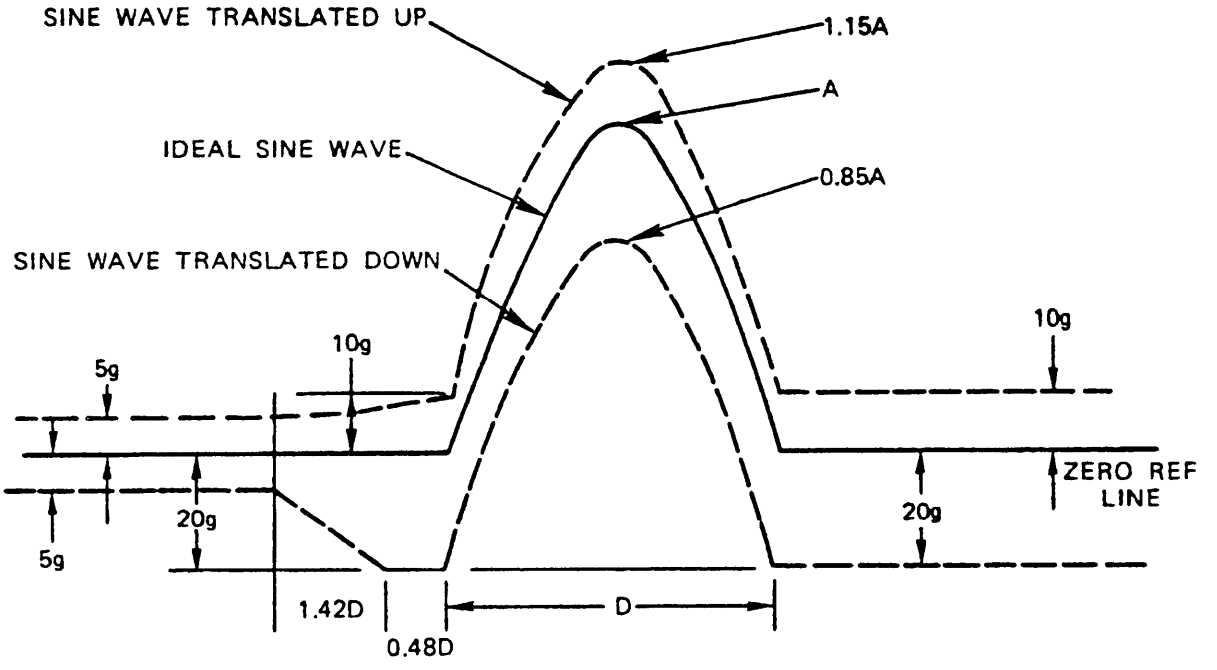


FIGURE 3. Acceleration sensitivity.

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NOTE:

The oscillogram shall include a time about 3D long with a pulse located approximately in the center. The acceleration amplitude of the ideal half sine pulse is A and its duration is D. The measured acceleration pulse shall be contained between the broken line boundaries.

FIGURE 4. Shock test.

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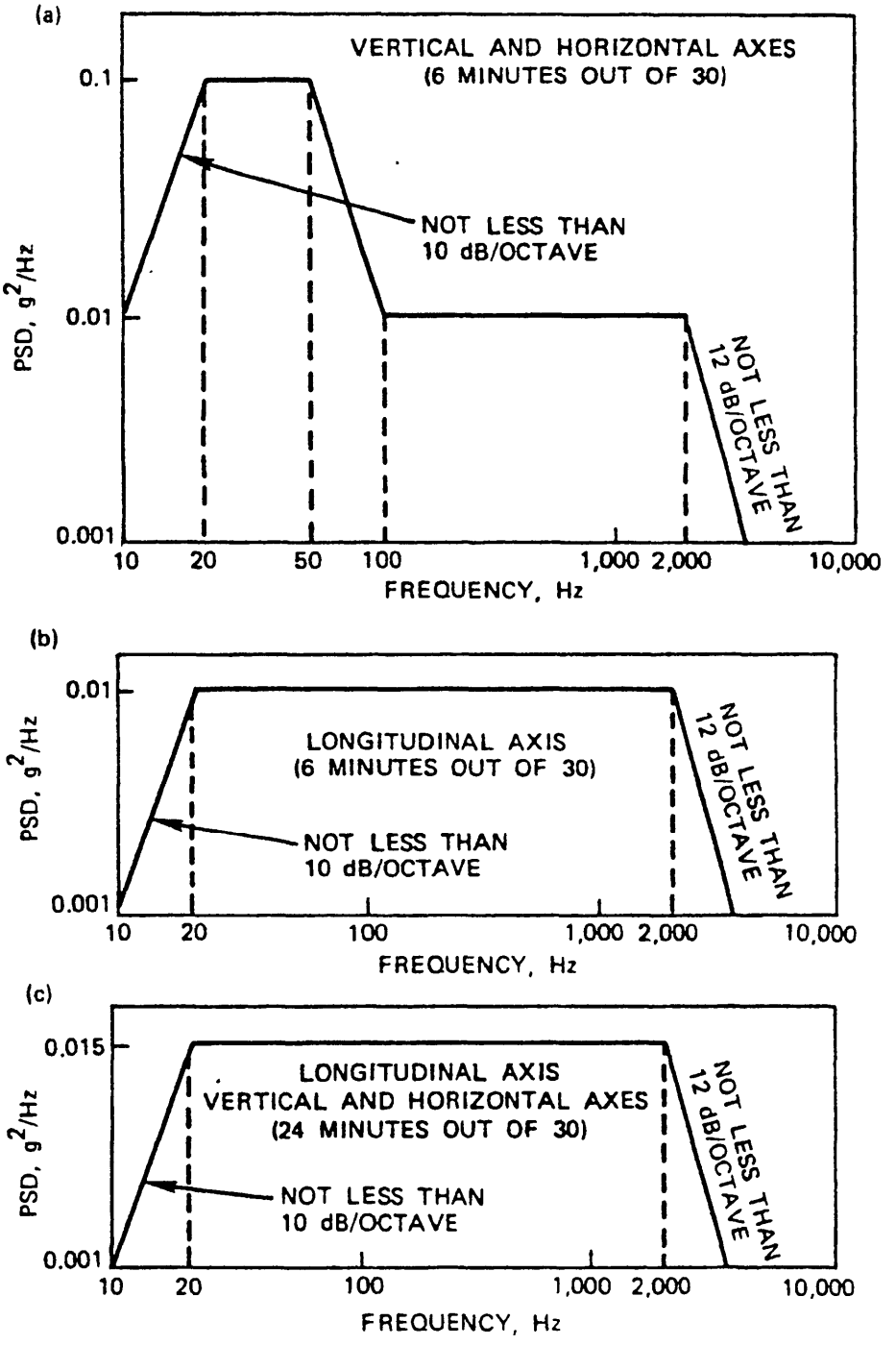


FIGURE 5. Vibration (random).

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NOTES:

1. A performance test shall be made every 6 ±3 cycles.
2. The vibration profile for the captive carry environment shall be as follows:

Vibration Duration (minutes per hour)	Input PSD Level (g^2/Hz)	Frequency (Hz)
50	0 005	20 to 2000
1 min 20 sec	0 015	20 to 2000
30 sec	0.05	20 to 100

   Input vibration level shall be monitored on the vibration fixture. Transition from one vibration level to another vibration level shall be made with minimal transitory time.
3. The total test time shall be 1800 hours, with no servo being subjected to less than one-half the average test time.

FIGURE 6. Captive carry environment profile.

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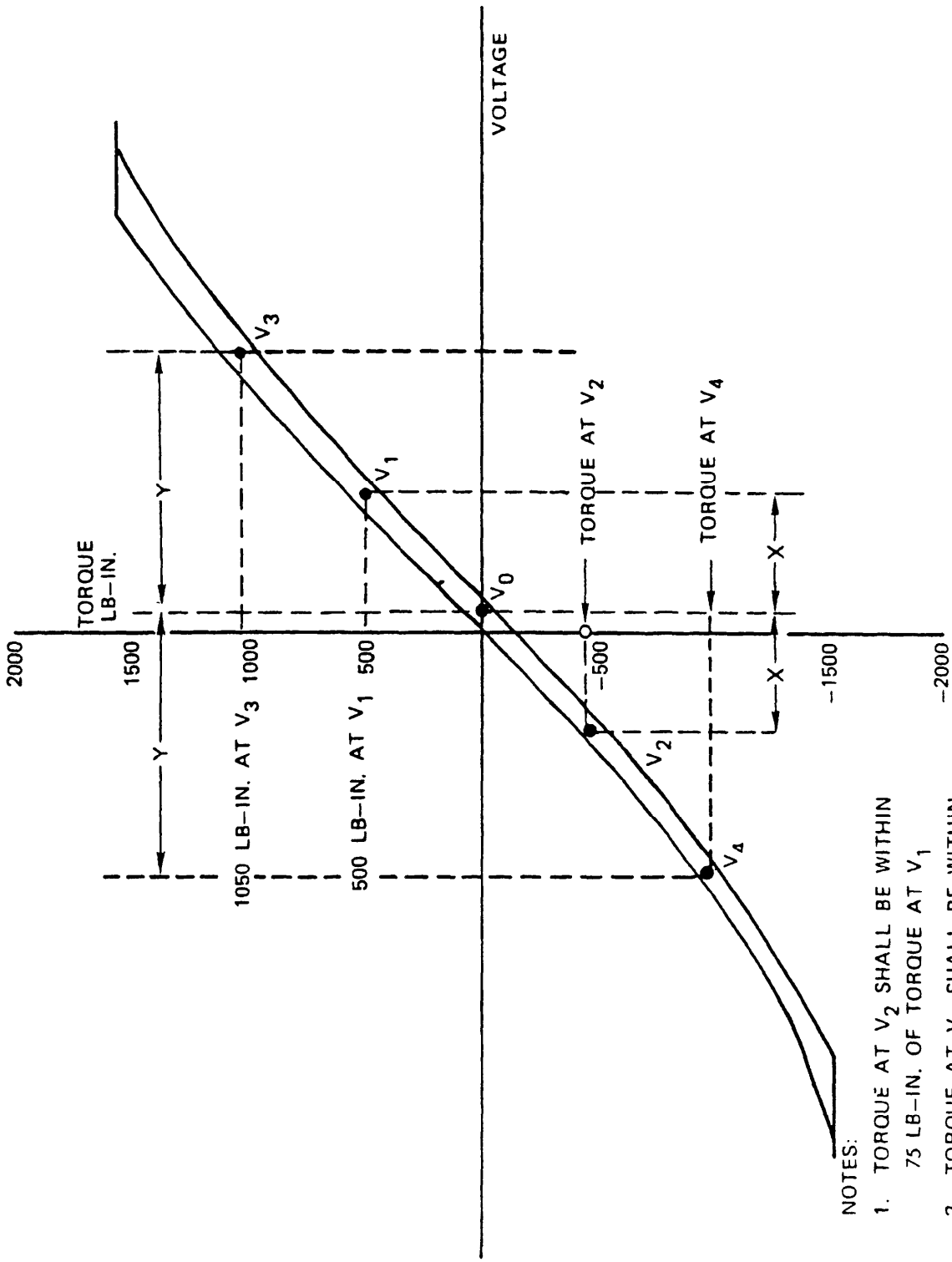


FIGURE 7. Torque symmetry.

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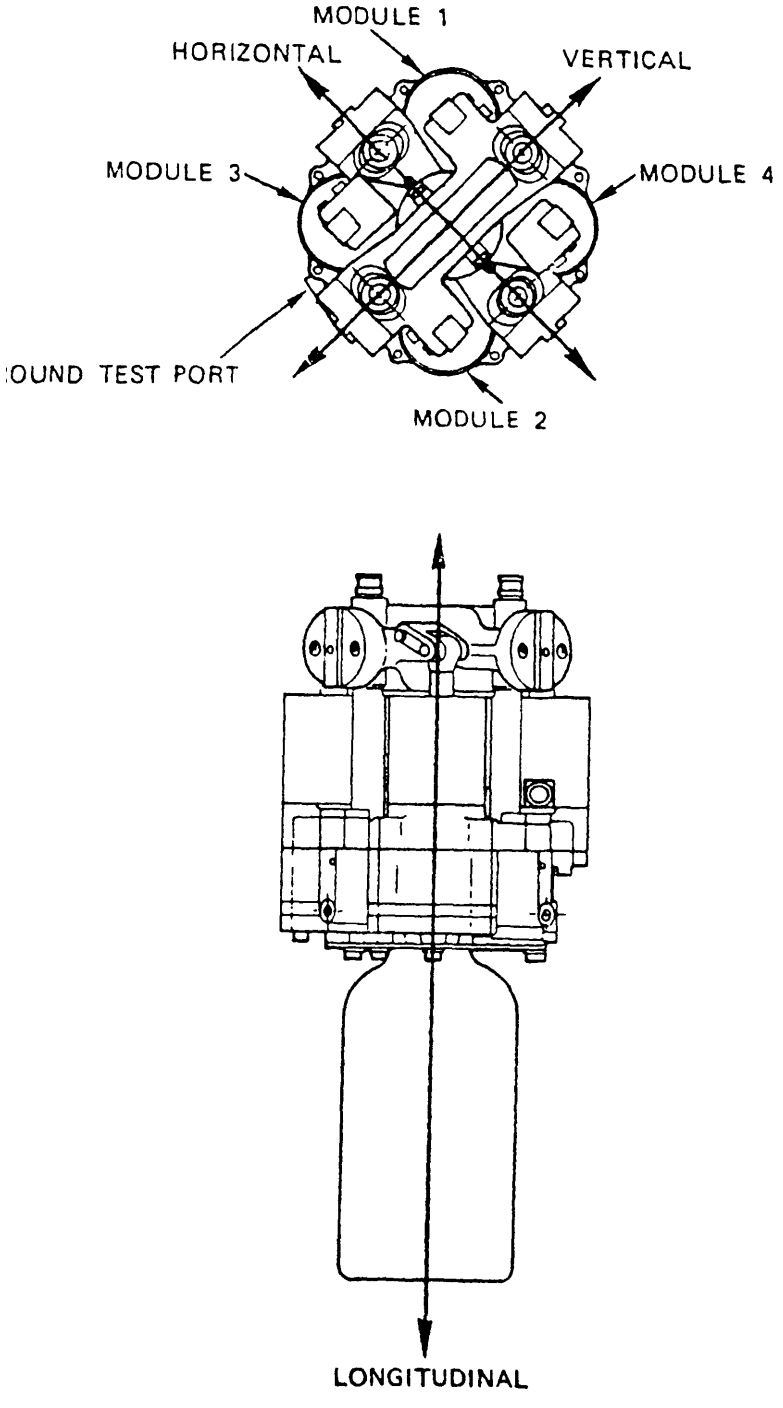


Figure 8. Vibration axes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-85565(AS)	MILITARY SPECIFICATION FOR SERVO, COLD GAS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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